

AI Overshoot: A Concrete Example of the Closure Overshoot Law

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Abstract

The global AI industry is often described as a “bubble” analogous to the dot-com or housing bubbles. This interpretation is incorrect and dangerously misleading. Using the Closure Overshoot Law from the Triadic Cosmos ecosystem, this paper demonstrates that AI is not experiencing a temporary valuation anomaly but a multi-domain overshoot across resource, energetic, entropic, geometric, and temporal dimensions. We analyze several interacting overshoot subsystems and show how their combined dynamics reduce the viable lifespan of the AI ecosystem, creating irreversible instability across economic, infrastructural, cognitive, industrial, and social closures. Unlike classical bubbles, the AI collapse would be systemic rather than financial.

1 The Closure Overshoot Law

Across physics, cosmology, ecology, strategy, cognition, and AGI, the Triadic Cosmos ecosystem reveals one universal architectural rule:

Any system that exceeds the limits of its local closure enters irreversible instability and collapses.

A local closure is a finite domain with its own causal structure, energetic budget, entropy dynamics, time flow, and corrective capacity. Examples include planets, economies, biospheres, submarines, and even stars.

Overshoot can take several forms:

- resource overshoot,
- energetic overshoot,
- entropic overshoot,
- geometric overshoot,
- temporal overshoot.

Once overshoot begins, corrective capacity falls below instability, time flow accelerates, entropy rises, and collapse becomes unavoidable.

2 What a Bubble Is

A bubble is a temporary mispricing of an asset. Classical bubbles share four properties:

1. the underlying asset has little or no intrinsic value,
2. valuation rises due to speculative dynamics,
3. the bubble pops when confidence collapses,
4. the system resets without structural collapse.

Dot-com stocks and housing derivatives fit this pattern: the collapse destroys value and can trigger a temporary financial crisis with an associated economic downturn, but it does not destabilize the underlying economic or infrastructural closure. The system eventually re-stabilizes because the collapse is financial rather than systemic.

3 Why AI Is Not a Bubble

The AI industry does not satisfy the defining criteria of a bubble:

- AI systems have real economic value and high adoption,
- demand is structural rather than speculative,
- the underlying technology works and is widely deployed,
- collapse would destabilize multiple closures simultaneously.

The AI ecosystem is not mispriced; it is *overstressed*. Its instability does not come from valuation but from overshoot across multiple finite domains.

4 AI as a Multi-Domain Closure Overshoot

The AI industry forms a closed ecosystem with finite energetic, material, entropic, cognitive, industrial, and temporal budgets. Expansionism—in the form of frontier scaling, hardware scaling, datacenter scaling, labor automation, and electrification—accelerates internal entropy production and reduces the viable lifespan of the closure.

We identify the following interacting overshoot subsystems.

4.1 Frontier AI Scaling

Goal: reach AGI or SI performance through model scaling.

Overshoot: diminishing returns, data saturation, rising inference costs, shorter model cycles, and increasing architectural opacity.

Collapse vector: temporal overshoot (accelerating model cycles) + entropic overshoot (opaque and unstable architectures).

Result: frontier scaling becomes economically and cognitively unsustainable, reducing the corrective capacity of the AI ecosystem.

4.2 Quantum Scaling

Goal: solve extremely complex problems via qubit scaling.

Overshoot: extreme cooling requirements, exotic materials, limited usable qubits, and narrow applicability.

Collapse vector: energetic overshoot (cryogenic demand) + geometric overshoot (specialized, non-integrable pipelines).

Result: quantum scaling fails to provide a viable alternative to classical AI and drains capital from the broader ecosystem.

4.3 Hardware Scaling

Goal: increase compute performance and efficiency.

Overshoot: HBM and storage scarcity, interconnect bottlenecks, cooling limits, vendor lock-in, and rapid hardware obsolescence.

Collapse vector: energetic overshoot (power and cooling saturation) + material overshoot (silicon, HBM, and supply chain constraints).

Result: hardware becomes the primary bottleneck for AI expansion, forcing higher costs and accelerating temporal pressure.

4.4 Datacenter Scaling

Goal: accommodate rising training and inference demand.

Overshoot: land, water, power, noise, long-term capital exposure, and hardware churn.

Collapse vector: energetic overshoot (grid saturation) + geometric overshoot (regional constraints) + resource overshoot (financial strain from capital intensity, rising interest rates, and rapid hardware obsolescence).

Result: datacenters become economically and energetically unstable, reducing the viability of large-scale AI deployment.

4.5 Space Datacenter Scaling

Goal: bypass terrestrial energy and cooling limits.

Overshoot: orbital scarcity, debris risk, maintenance impossibility, and extreme resource waste.

Collapse vector: geometric overshoot (orbital congestion).

Result: space-based datacenters prove non-viable and divert capital and engineering capacity from terrestrial stability.

4.6 Fusion Energy Scaling

Goal: solve AI energy demand with fusion.

Overshoot: rising prototype complexity, uncertain viability, and escalating capital drain.

Collapse vector: energetic overshoot (unbounded energy demand).

Result: fusion fails to stabilize AI energy requirements and becomes a long-term financial sink.

4.7 Software Engineer Pool Downscaling

Goal: replace human engineers with agentic labor.

Overshoot: unstable codebases, loss of junior pipeline, rising senior pressure, and declining talent availability.

Collapse vector: entropic overshoot (cognitive and architectural instability).

Result: software ecosystems degrade, reducing reliability and increasing long-term maintenance costs.

4.8 Labor Market Downscaling

Goal: automate human labor.

Overshoot: reduced diversity, unpredictable safety failures, declining consumer base, and rising liabilities.

Collapse vector: social-economic overshoot (demand collapse).

Result: automation reduces economic resilience and shrinks the consumer market needed to sustain AI services.

4.9 Advertising Scaling Overshoot

Goal: subsidize AI inference and platform costs through advertising revenue.

Overshoot: global advertising budgets form a closed economic domain that does not scale with the number of competing platforms. Declining consumer purchasing power and inflation further reduce real advertising value.

Collapse vector: entropic overshoot (platform fragmentation) + energetic overshoot (attempting to fund rising inference costs with a fixed budget) + temporal overshoot (model cycles outpacing revenue growth).

Result: advertising fails as a sustainable funding mechanism and accelerates collapse by reducing economic corrective capacity.

4.10 Automotive Scaling Overshoot

Goal: scale the automotive industry through electrification, autonomy, and software-defined vehicles.

Overshoot: rising demand for batteries, rare metals, and chips; increasing software complexity; higher consumer financing pressure; rapid EV depreciation; and global market stress.

Collapse vector: energetic overshoot (battery production and grid stress) + material overshoot (rare metal and chip scarcity) + entropic overshoot (software-complex vehicles) + temporal overshoot (accelerating depreciation and financing pressure).

Result: automotive instability weakens supply chains, reduces consumer demand, and amplifies pressures on the AI ecosystem.

4.11 AI Dependence Overshoot

Goal: integrate AI systems deeply into education, business processes, professional workflows, and everyday consumer activity.

Overshoot: rising societal dependence on continuous AI availability and reliability. Children learn AI-assisted methods in school, companies embed AI into core operations, and individuals rely on AI for communication, information retrieval, planning, and decision support. As AI becomes a cognitive and infrastructural backbone, temporary disruptions or large-scale instability can propagate rapidly across society.

Collapse vector: social-cognitive overshoot (loss of functional capacity when AI systems degrade or fail).

Result: although recoverable in principle—human societies have lived without AI—the sudden

loss or degradation of AI services would trigger acute crises across education, business continuity, logistics, administration, and public communication. AI dependence therefore amplifies the societal impact of any instability arising from the broader overshoot dynamics.

4.12 Governance and Regulatory Overshoot

Goal: develop and implement effective governance, regulatory frameworks, and oversight mechanisms that ensure safe, stable, and sustainable AI deployment.

Overshoot: the pace of AI scaling exceeds the capacity of governments, institutions, and regulatory bodies to respond. Regulatory development requires time, expertise, funding, political coordination, and cross-sector alignment, all of which are limited resources. As AI systems accelerate in complexity, scale, and societal impact, governance structures lag behind, creating a growing gap between technological capability and institutional corrective capacity. Economic pressures, strategic competition, and fragmented jurisdictional interests further slow regulatory progress.

Collapse vector: temporal overshoot (regulation cannot keep pace with model cycles), resource overshoot (insufficient institutional capacity, funding, and expertise), and entropic overshoot (fragmented, inconsistent, or contradictory governance across regions).

Result: governance becomes a structural bottleneck. Without a slowdown or freeze, regulatory systems cannot stabilize the ecosystem before overshoot dynamics trigger broader collapse. Governance overshoot therefore amplifies systemic risk by reducing society’s ability to implement timely corrective measures.

4.13 Debt and Energy Crisis (Accelerator)

Goal: finance expansion under rising costs.

Overshoot: interest rate pressure, energy price inflation, and capital scarcity.

Collapse vector: temporal overshoot (accelerating financial decay).

Result: debt and energy dynamics amplify all other overshoot vectors, accelerating collapse across the entire AI ecosystem.

5 Helicopter View: The AI Collapse Clock

These combined overshoot domains form a self-reinforcing loop:

1. scaling increases resource demand,
2. resource demand increases cost,
3. cost accelerates temporal pressure,
4. temporal pressure increases complexity,
5. complexity increases entropic load,
6. entropic load reduces corrective capacity,
7. reduced corrective capacity accelerates collapse.

This dynamic is not speculative but structural. It matches the Closure Overshoot Law exactly: expansionism shortens the lifespan of a closed world.

6 Conclusion

AI is not a bubble. It is a multi-domain closure overshoot whose collapse would destabilize economic, infrastructural, cognitive, industrial, and social systems. The industry’s instability arises not from mispricing but from exceeding the limits of its local closure. As with all closed ecosystems, respecting those limits is the only path to long-term viability.

This analysis also clarifies why a voluntary slowdown or freeze in AI scaling is not merely a safety measure aimed at preventing misuse or uncontrolled autonomy. It is a structural intervention that restores stability to the broader ecosystem. By reducing temporal pressure, resource consumption, energetic load, and entropic accumulation, a freeze increases the corrective capacity of the system and prevents the self-reinforcing overshoot dynamics that otherwise lead to systemic collapse. In this sense, a freeze is not an ideological stance but an architectural necessity for long-term economic and technological resilience.

Moreover, unlike scenarios that would require rapid technological expansion to address external existential pressures, there is no external driver that necessitates accelerated AI growth. The forces pushing AI beyond the limits of its local closure are entirely internal to the economic, industrial, and institutional ecosystem. This absence of external urgency further strengthens the case for a voluntary freeze as a rational stabilizing measure: without external threats demanding acceleration, slowing down becomes the only structurally coherent response to overshoot.

A Triadic Cosmos Ecosystem Context

The present paper is part of the broader *Triadic Cosmos* ecosystem: a unified research program that develops a compositional ontology of intelligence together with modular, inspectable computational architectures. The ecosystem spans conceptual theory, executable reference implementations, and fully documented empirical demonstrations, forming a coherent framework for studying hybrid intelligent systems.

All materials, extended notes, and evolving documentation are publicly available in the open GitHub project:

<https://github.com/triadic-cosmos>